Sultanate of Oman
Ministry of Agriculture, Fisheries & Water Resources

# Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate


# Wadi Majlas Flood Protection Dam
## Hydrology Report

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# EXECUTIVE SUMMARY

This report presents the updated hydrological study for the **Wadi Majlas Flood Protection Dam**, Sultanate of Oman. It supersedes the hydrology chapter of the previous design report and documents the rainfall data analysis, gap filling, regional frequency analysis, Probable Maximum Precipitation (PMP), rainfall-runoff modelling, calibration review and design flood estimates for the dam and the downstream dyked-channel protection scheme.

The catchment draining to the dam is about **571 km2** (604 km2 including the downstream sub-catchments added for the dyked-channel design), rising from ~80 m at the dam to nearly 1,970 m in the Hajar mountains. A HEC-HMS model was built on a 5 m NSA DTM with an SCS curve-number loss method (GCN250 grid, catchment average CN about 89), Kirpich lag times and depth-area reduced 24-hour design storms.

Design rainfall was derived by a **regional frequency analysis following the Hosking & Wallis index-flood procedure with the GEV distribution fitted by L-moments**. Annual maximum series (the maximum annual rolling sum for each duration) were extracted on a water-year basis from nine gauges, after 3D inverse-distance gap-filling of the 10-minute records; years in which a gauge was offline or clearly missed the dominant regional storm were excluded. Five gauges with at least ten complete water years form a single, statistically **homogeneous** region (heterogeneity H1 = -1.6; no discordant sites). Because Oman rainfall is moisture-limited (24-hour PMP about 1,300 mm), the GEV shape parameter was constrained to the physically bounded case (kappa >= 0), which keeps the extreme quantiles plausible. The resulting local 24-hour depths are about 1.5 times the Oman Flood-Mapping design rainfall at moderate return periods and converge with it at rare return periods; the Flood-Mapping depths, which drove the HEC-HMS runs, were retained as the design basis.

A review of the previous calibration found that the observed Gonu hydrograph at the Majlass gauge could only be reproduced by reducing gauged rainfall to physically implausible levels, whereas the Hayfadh gauge was well reproduced. That observed hydrograph is judged erroneous and was excluded; the previous report did not flag it.

Design peak discharges were produced for return periods from 2 to 10,000 years plus the PMP, with and without the Majlas reservoir. The reservoir stores the full inflow up to about the 200-year event and attenuates the 10,000-year peak from about 14,170 to 8,750 m3/s (~38%).

# 1. INTRODUCTION

The Ministry of Agriculture, Fisheries and Water Resources is developing a flood-protection scheme for Wadi Majlas, in Wilayat Qurayat (Muscat Governorate), comprising a flood-detention dam and a downstream dyked (leveed) channel. This report updates the hydrological basis of the scheme. Its objectives are to:

- characterise the Wadi Majlas catchment and its hydro-meteorological setting;
- analyse the available rainfall records (quality control, gap filling) and derive design rainfall for durations to 24 hours and return periods to 10,000 years, and estimate the PMP;
- build and calibrate a HEC-HMS rainfall-runoff model of the catchment and downstream reaches;
- review the calibration adopted in the previous study against the observed cyclone hydrographs; and
- produce design flood hydrographs and peak discharges for the dam and the downstream protection works.

Two HEC-HMS models were developed: an initial model used to construct and calibrate the catchment response against observed hydrographs and peak flows, and a final model that adds the downstream sub-catchments for the dyked-channel design and applies a depth-area analysis to represent the areal reduction of extreme rainfall.

# 2. STUDY AREA AND PHYSIOGRAPHY

The Wadi Majlas catchment drains the eastern Hajar mountains towards the dam site; the contributing area to the dam is **571 km2**. Elevations range from about 80 m a.s.l. at the dam to nearly 1,970 m at the highest divide, giving a strongly mountainous, high-relief catchment with a dense, steep drainage network (Figure 1). Physiographic parameters derived from the 5 m NSA DTM are given in Table 1.

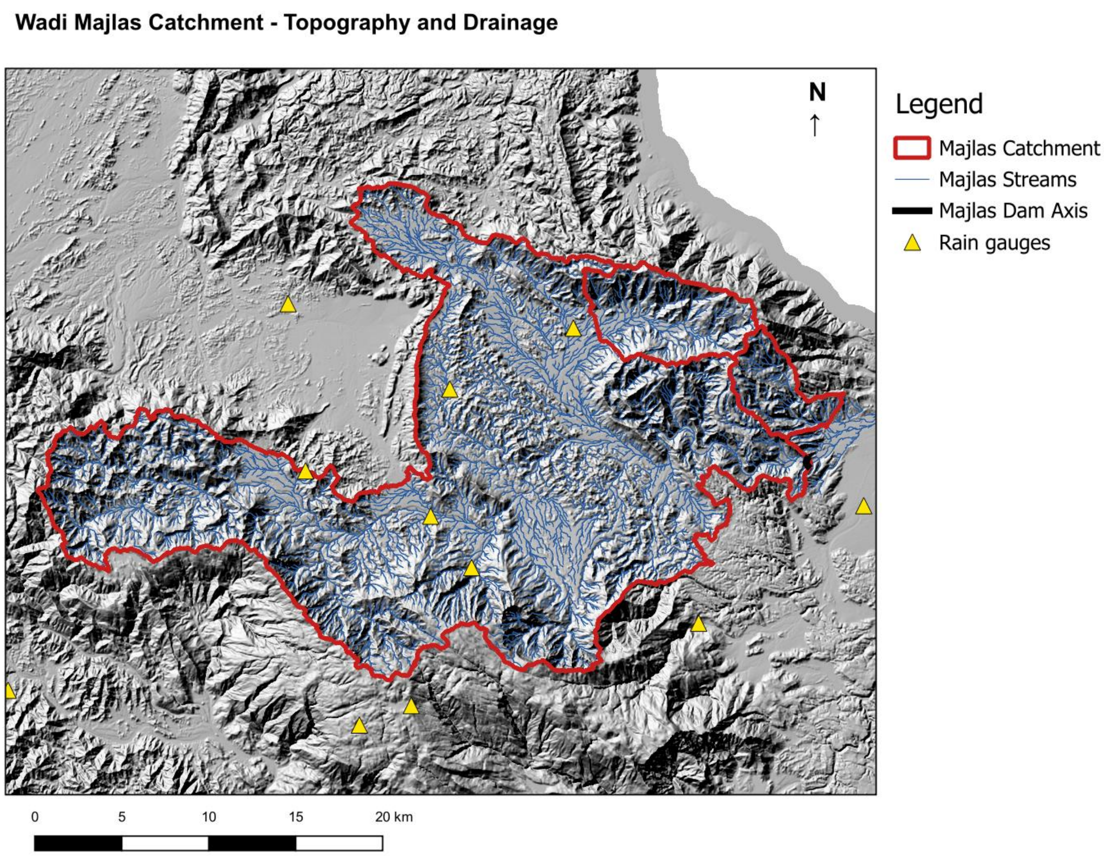


**Figure 1 Wadi Majlas catchment - topography, drainage network and rain gauges (QGIS; hillshade from DTM_5m).**

**Table 1 Physiographic characteristics of the Wadi Majlas catchment (DTM_5m).**

| Physiographic parameter | Value |
|---|---|
| Contributing area to dam | 571 km2 |
| Catchment-average curve number (CN) | 89 |
| Time of concentration, Tc | 400 min |
| Longest flow path length | 68.4 km |
| Longest flow-path slope | 0.0216 m/m |

| Physiographic parameter | Value |
|---|---|
| 10-85% flow-path length | 51.3 km |
| 10-85% flow-path slope | 0.0088 m/m |
| Mean basin slope | 0.546 m/m |
| Basin relief | 1,948 m |
| Elongation ratio | 0.394 |
| Drainage density | 0.161 km/km2 |

## 2.1. METEOROLOGY

The regional climate is hot arid, with intense but infrequent rainfall from tropical cyclones and convective storms. Monthly air-temperature statistics for the representative Nakhal station are shown in Figure 2 (mean 24-35 C, recorded maxima above 50 C). The wind regime at the dam site, from 20 years of ERA5-Land 10 m wind (2001-2021, Figure 3), is bimodal - drainage winds from the NNE and SSW - with a mean speed of about 1.8 m/s. These support the PMP moisture-maximisation and the reservoir evaporation assessment.

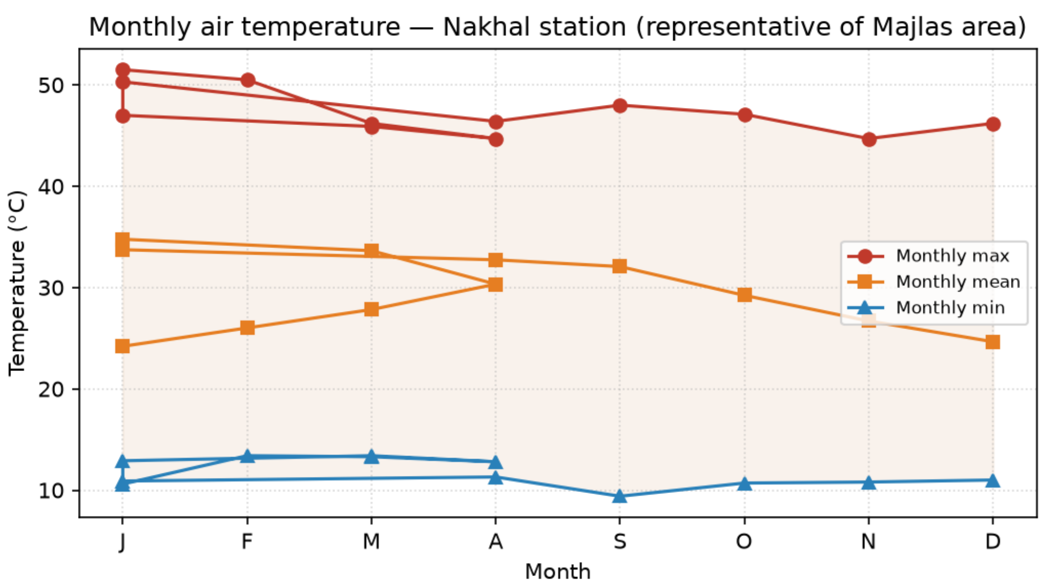


**Figure 2 Monthly air temperature at Nakhal station (representative of the Majlas area).**

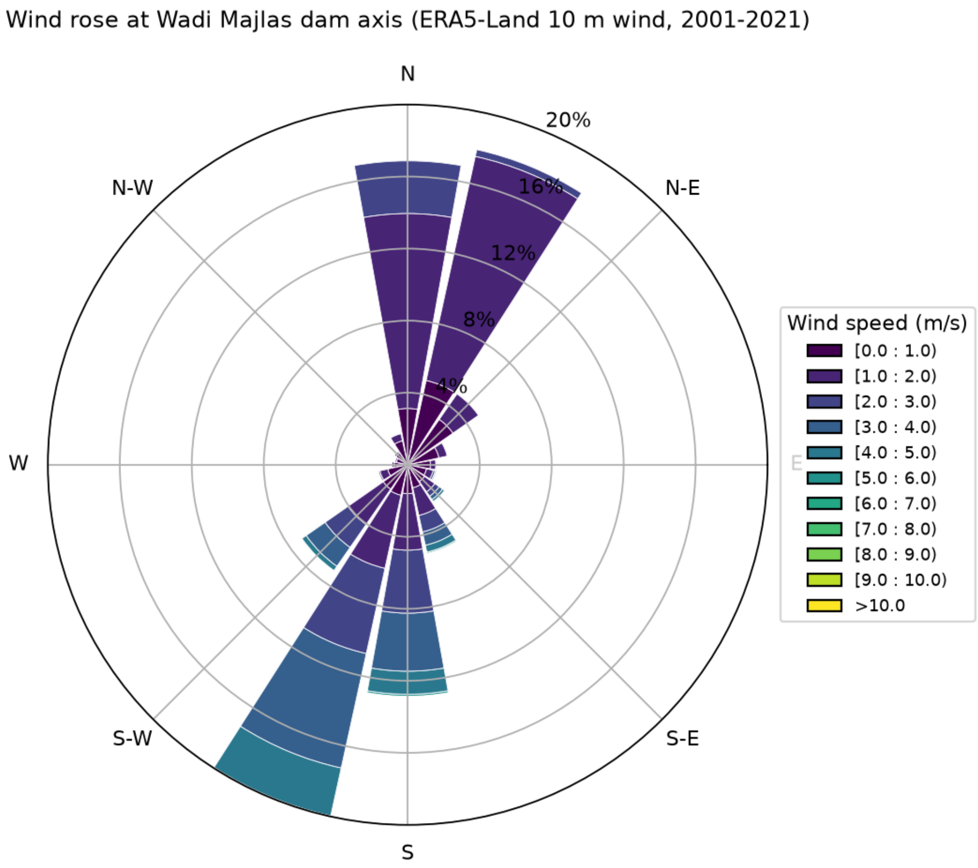


**Figure 3 Wind rose at the Wadi Majlas dam axis (ERA5-Land 10 m wind, 2001-2021).**

# 3. AVAILABLE DATA AND RAINFALL RECORDS

The analysis draws on sub-daily (10-minute) rainfall from nine gauges in and around the catchment, the Oman Flood-Mapping regional design rainfall, the 5 m NSA DTM, the GCN250 curve-number grid, reservoir drone-survey hypsometry, and observed cyclone hydrographs (Gonu 2007, Phet 2010). The gauge inventory and cleaned 24-hour annual-maximum statistics are given in Table 2; records span 1994-2024, with the in-catchment (Majlas) gauges - Buei, Taba and Hayfadh - among the longest. Five gauges have at least ten complete water years and form the fitting set for frequency analysis.

**Table 2 Rain-gauge inventory and cleaned 24-hour annual-maximum statistics.**

| Gauge | Elev (m) | 24-h AMS (yr) | Mean (mm) | Median (mm) | Max (mm) | In fit |
|---|---|---|---|---|---|---|
| Jabal Abyadh East | 1690 | 13 | 111 | 62 | 625 | yes |
| Jabal Al Aswad | 0 | 1 | 139 | 139 | 139 | no |
| Sarin near Habubiyah | 445 | 3 | 78 | 68 | 137 | no |
| Buei near Buei | 450 | 18 | 56 | 22 | 379 | yes |
| Siya at Siya | 371 | 6 | 28 | 19 | 75 | no |
| Taba at Taba | 310 | 18 | 65 | 28 | 518 | yes |
| Hayfadh at Hayfadh | 230 | 17 | 65 | 24 | 621 | yes |
| Qurayat at Qurayat | 20 | 15 | 52 | 27 | 229 | yes |
| Abayah near Misfah | 623 | 6 | 48 | 36 | 101 | no |

# 4. STATISTICAL ANALYSIS OF RAINFALL DATA

## 4.1. ANNUAL MAXIMUM SERIES AND DATA QUALITY

For each duration from 5 minutes to 96 hours, the **annual maximum series (AMS) is the maximum annual rolling sum** of rainfall: the 10-minute record is accumulated over a moving window equal to the duration, and the largest value in each water year (1 October - 30 September, to avoid splitting the June cyclones) is taken. A water year is retained only if the gauge was operational; years in which a gauge was offline (no record) or clearly missed the year's dominant regional storm were excluded, as these represent missing data rather than low annual maxima. Each 24-hour series was tested for outliers, trend and change-point (Table 3). The 2007 Cyclone Gonu maxima (500-625 mm) are flagged by the Grubbs test but are **retained** as genuine extreme observations; no significant Mann-Kendall trend or Pettitt change-point was found, so the series are treated as stationary.

**Table 3 Outlier (Grubbs), trend (Mann-Kendall) and change-point (Pettitt) tests, 24-hour AMS.**

| Gauge | N | Grubbs (largest value) | MK trend | Pettitt change-point |
|---|---|---|---|---|
| Jabal Abyadh East | 13 | 625 (cyclone-retained) | none | stationary |
| Buei near Buei | 18 | 379 (cyclone-retained) | none | stationary |
| Taba at Taba | 18 | 518 (cyclone-retained) | none | stationary |
| Hayfadh at Hayfadh | 17 | 621 (cyclone-retained) | none | stationary |
| Qurayat at Qurayat | 15 | 229 (ok) | none | stationary |

## 4.2. GAP FILLING

Genuine gaps in the 10-minute records (timestamps within a gauge's operational period that were not observed) were infilled by **three-dimensional inverse-distance weighting (IDW)** from neighbouring gauges, with weight 1/d^2 over a horizontal search radius of 30 km and a vertical term (lambda = 0.005) to account for the strong orographic gradient:

$$P_t = \frac{\sum_i w_i P_{i,t}}{\sum_i w_i}, \quad w_i = 1/d^2$$

A gap is filled only when a storm was actually active: at least one donor within the radius has a 24-hour rolling total above 10 mm, and - the key control adopted here - **at least three donors recorded non-zero rainfall in that same 10-minute step**. Three rules prevent the errors of the earlier analysis:

the infilling is a **single pass using observed values only** (filled values are never re-used as donors); a **recorded zero is a real zero, not a gap**; and an existing observed value is never overwritten. The two cyclone windows (Gonu, Phet) are excluded from filling to avoid inflating the extremes.

## 4.3. REGIONAL FREQUENCY ANALYSIS (HOSKING & WALLIS)

Design rainfall was estimated by the **index-flood regional frequency analysis of Hosking & Wallis (1997)**. Each gauge AMS is standardised by its at-site mean (the index rainfall) and pooled into a single regional dimensionless growth curve, increasing the effective record length and the reliability of rare quantiles. The five fitting gauges (81 pooled station-years) form **one region**. Discordancy is below the critical value for every gauge (Table 4), and the heterogeneity measure confirms a statistically **homogeneous** region for all design-relevant durations (24-hour H1 = -1.6; H1 < 1 for 30 min and longer).

**Table 4 At-site 24-hour L-moment ratios and Hosking & Wallis discordancy Di (critical Di = 3; region homogeneous, H1 = -1.6).**

| Gauge | N | Mean (mm) | L-CV | L-skew | L-kurt | Di |
|---|---|---|---|---|---|---|
| Jabal Abyadh East | 13 | 111.1 | 0.639 | 0.602 | 0.516 | 1.2 |
| Buei near Buei | 18 | 56.2 | 0.673 | 0.687 | 0.569 | 1.31 |
| Taba at Taba | 18 | 64.9 | 0.681 | 0.728 | 0.652 | 0.28 |
| Hayfadh at Hayfadh | 17 | 65.2 | 0.724 | 0.814 | 0.752 | 0.9 |
| Qurayat at Qurayat | 15 | 51.5 | 0.59 | 0.62 | 0.467 | 1.31 |

## 4.4. DISTRIBUTION AND DESIGN DEPTHS

The pooled data are fitted with the **Generalised Extreme Value (GEV) distribution by L-moments**. On this short record the 24-hour L-skewness is high (about 0.7) because a single event - Cyclone Gonu (2007) - is far rarer than the ~20-year record; the unconstrained L-moment fit therefore returns a negative shape parameter (kappa < 0), an unbounded tail that extrapolates to physically impossible depths (over 10,000 mm). Because Oman rainfall is **moisture-limited** - the atmosphere over the Arabian Sea can deliver only a finite depth, quantified by the 24-hour PMP of about 1,300 mm - the tail must be bounded. The GEV shape was therefore constrained to the physically admissible bounded case (kappa >= 0), keeping the location and scale from the L-moments. At short durations the L-moments already give kappa >= 0, so those curves are unchanged; only the longduration tails are bounded. The resulting regional growth curve (Figure 4) gives the 24-hour design depths in Table 5; the 10,000-year depth (about 580 mm) sits comfortably below the PMP.

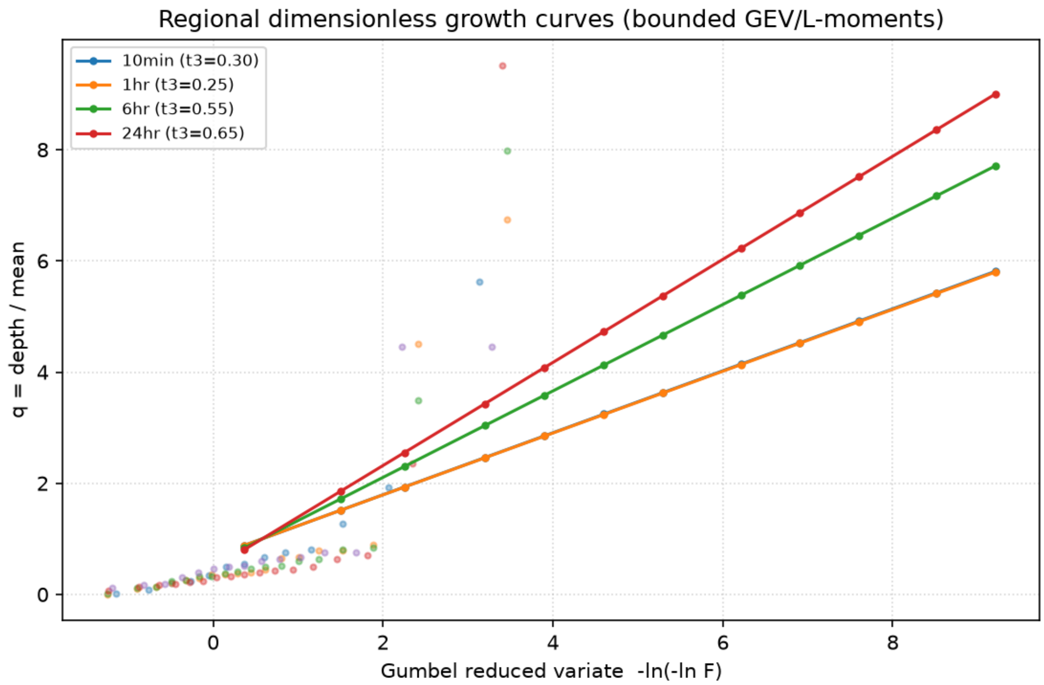


**Figure 4 Regional dimensionless growth curves (GEV/L-moments, bounded shape kappa >= 0).**

**Table 5 Regional 24-hour design rainfall depths (local index-flood RFA, bounded GEV/L-moments).**

| Return period (yr) | 24-h design depth (mm) |
|---|---|
| 2 | 52 |
| 5 | 120 |
| 10 | 166 |
| 25 | 223 |
| 50 | 265 |
| 100 | 307 |
| 200 | 349 |
| 500 | 404 |
| 1000 | 446 |
| 2000 | 488 |
| 5000 | 543 |
| 10000 | 584 |

## 4.5. INTENSITY-DURATION-FREQUENCY (IDF) CURVES

Station IDF curves (index flood x regional growth curve, per gauge) and a single regional IDF from the pooled data were derived for all durations from 5 minutes to 96 hours. Depths are enforced non-decreasing with duration, and the bounded shape keeps the curves physically plausible and free of the post-hour crossing seen when durations are fitted independently. The station IDF curves are shown in Figure 5 and the regional IDF in Figure 6.

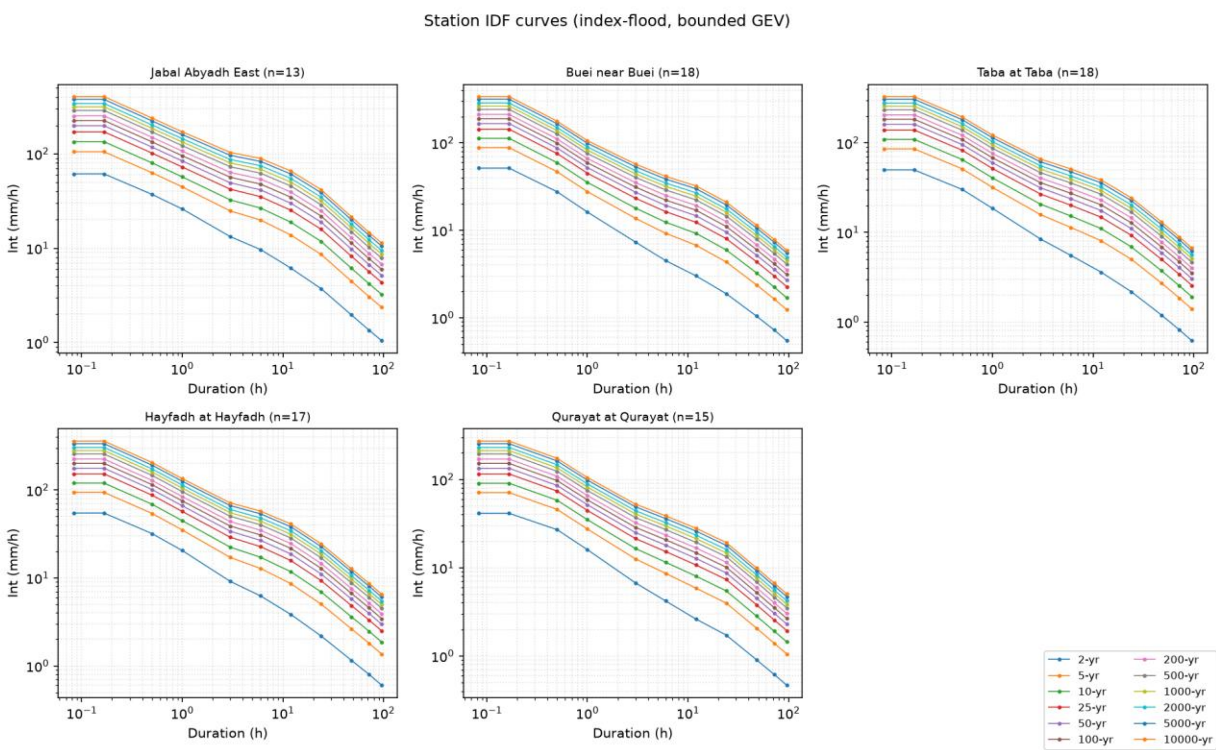


**Figure 5 Station IDF curves (index-flood, bounded GEV/L-moments).**

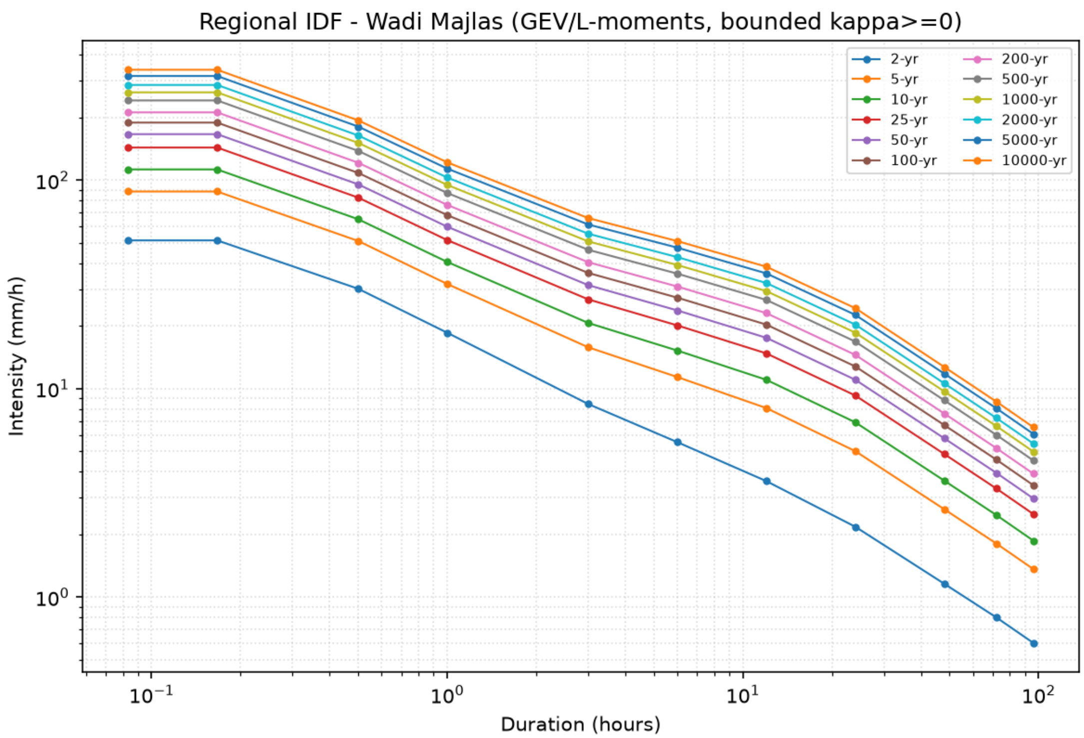


**Figure 6 Regional IDF curves for the Wadi Majlas study area (bounded GEV/L-moments).**

**Table 6 Regional design rainfall depth (mm) by duration and return period (bounded GEV/L-moments).**

| Duration | 2-yr | 10-yr | 50-yr | 100-yr | 1,000-yr | 10,000-yr |
|---|---|---|---|---|---|---|
| 5 min | 4 | 9 | 14 | 16 | 22 | 28 |
| 10 min | 9 | 19 | 28 | 32 | 44 | 57 |
| 30 min | 15 | 32 | 48 | 54 | 76 | 97 |
| 1 h | 19 | 41 | 60 | 68 | 95 | 122 |
| 3 h | 25 | 62 | 94 | 108 | 153 | 198 |
| 6 h | 33 | 92 | 142 | 164 | 235 | 307 |
| 12 h | 43 | 132 | 211 | 244 | 353 | 462 |
| 24 h | 52 | 166 | 265 | 307 | 446 | 584 |
| 48 h | 56 | 174 | 277 | 321 | 466 | 610 |
| 72 h | 58 | 178 | 284 | 329 | 476 | 624 |
| 96 h | 58 | 179 | 286 | 330 | 479 | 627 |

# 5. COMPARISON WITH THE OMAN FLOOD-MAPPING RAINFALL

The Oman Flood-Mapping study provides regional 24-hour design rainfall covering the Majlas catchment; these depths drove all HEC-HMS design runs. Table 7 and Figure 7 compare them with the local RFA depths.

**Table 7 24-hour design rainfall: local index-flood RFA versus Oman Flood-Mapping.**

| Return period (yr) | Local RFA (mm) | Flood Mapping (mm) | Ratio |
|---|---|---|---|
| 2 | 52 | 28 | 1.89 |
| 5 | 120 | 55 | 2.18 |
| 10 | 166 | 106 | 1.57 |
| 25 | 223 | 138 | 1.62 |
| 50 | 265 | 168 | 1.58 |
| 100 | 307 | 205 | 1.49 |
| 200 | 349 | 251 | 1.39 |
| 500 | 404 | 327 | 1.23 |
| 1000 | 446 | 400 | 1.11 |
| 2000 | 488 | 489 | 1.00 |
| 5000 | 543 | 639 | 0.85 |
| 10000 | 584 | 685 | 0.85 |

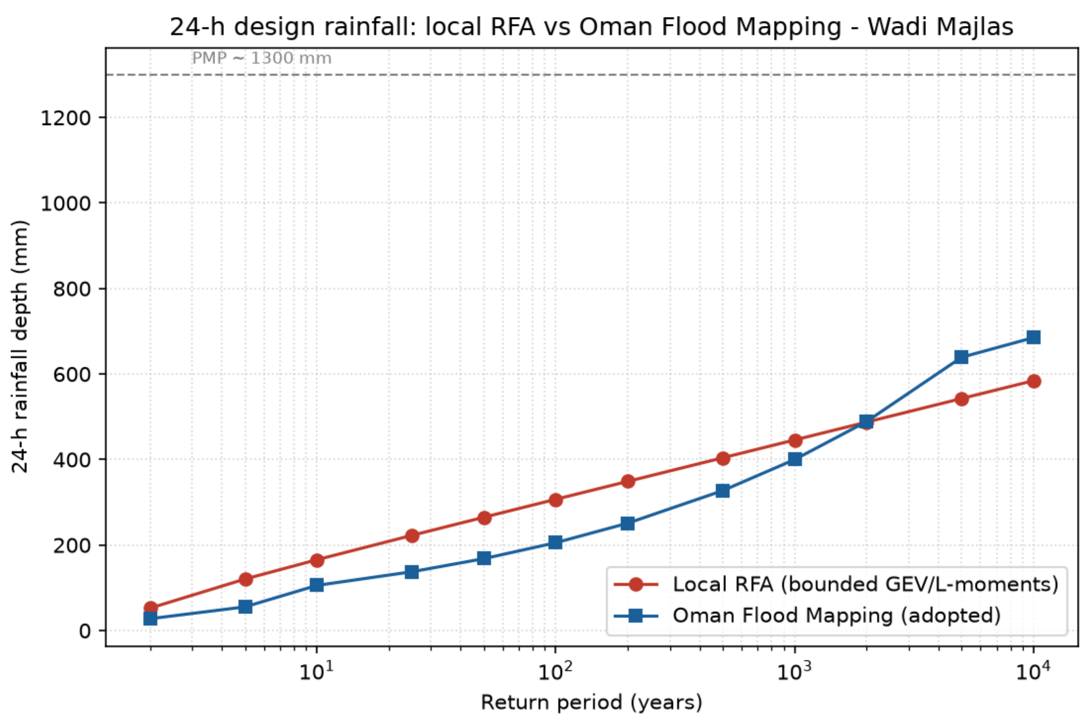

**Figure 7 24-hour design rainfall - local RFA versus Oman Flood-Mapping (both below the PMP).**

Both curves remain the same order of magnitude and well below the PMP. The local depths are about 1.5-1.9 times the Flood-Mapping values at low-to-moderate return periods and converge with them at rare return periods (within 15% beyond the 1,000-year event). The higher local values reflect the gauges directly capturing Cyclone Gonu (2007), whereas the Flood-Mapping product is a spatially smoothed regional grid. Because the Flood-Mapping rainfall draws on a longer, spatially extensive database and was already used to drive the HEC-HMS design runs, it is **retained as the design basis**; the local analysis is an independent check that confirms the same order of magnitude and quantifies the local orographic/Gonu enhancement.

# 6. PROBABLE MAXIMUM PRECIPITATION (PMP)

The PMP is the greatest depth of precipitation meteorologically possible for a given duration over the basin (WMO, 2009). The deterministic moisture-maximisation and storm-transposition approach of the regional Oman PMP study was adopted. Cyclone Gonu (2007), the highest recorded regional 24-hour rainfall, was maximised and transposed to the project area, the storm rainfall being scaled by the ratio of precipitable water under maximum and observed dew-point conditions:

$$PMP = R_0 \frac{W_m}{W_s}$$

ERA-5 dew-point temperatures (1940-present) provided the maximising factors. The resulting 24-hour PMP grid over the catchment (Figure 8) ranges from about 780 mm near the dam to over 1,600 mm on the highest ground, with a catchment-average of about **1,300 mm** - the physical upper bound used to constrain the rainfall-frequency tail in Section 4.

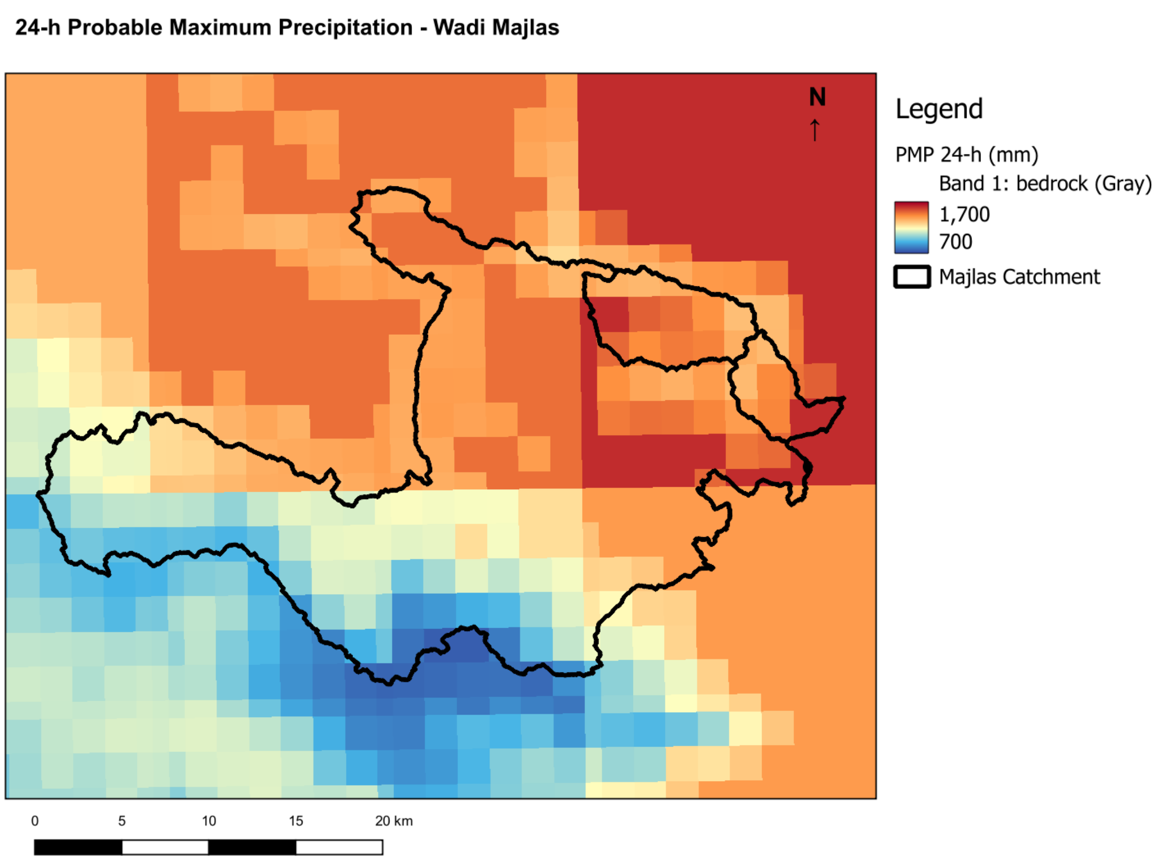


**Figure 8 24-hour Probable Maximum Precipitation over the Wadi Majlas catchment (QGIS).**

# 7. RAINFALL-RUNOFF MODEL

The catchment response was simulated in HEC-HMS with the SCS curve-number loss method, the SCS unit-hydrograph transform with Kirpich lag times and reach routing. Design storms were 24-hour hyetographs with the Flood-Mapping temporal distribution and a depth-area reduction factor.

## 7.1. SCS CURVE-NUMBER LOSS METHOD

Direct runoff is computed from the SCS relationship:

$$Q = \frac{(P - I_a)^2}{(P - I_a + S)}$$

where Q is runoff, P is rainfall, Ia is initial abstraction and S the potential maximum retention, with

$$S = \frac{25400}{CN} - 254 \ \ (mm) \qquad I_a = 0.2\,S$$

Arid- and semi-arid rangeland curve numbers follow the Oman Hydrological Design Standards (OHDS-2017); for the mountainous catchment the applicable values are CN 86-90 (slope < 18%) and 90-94 (slope > 18%).

## 7.2. CURVE NUMBER GRID

Curve numbers were taken from the GCN250 global product (250 m; ESA CCI land cover with HYSOGs250m soil groups) and clipped to the catchment (Figure 9). The catchment-average CN is about **89** (AMC II); sub-basin values range 86.5-91.0 (Table 9).

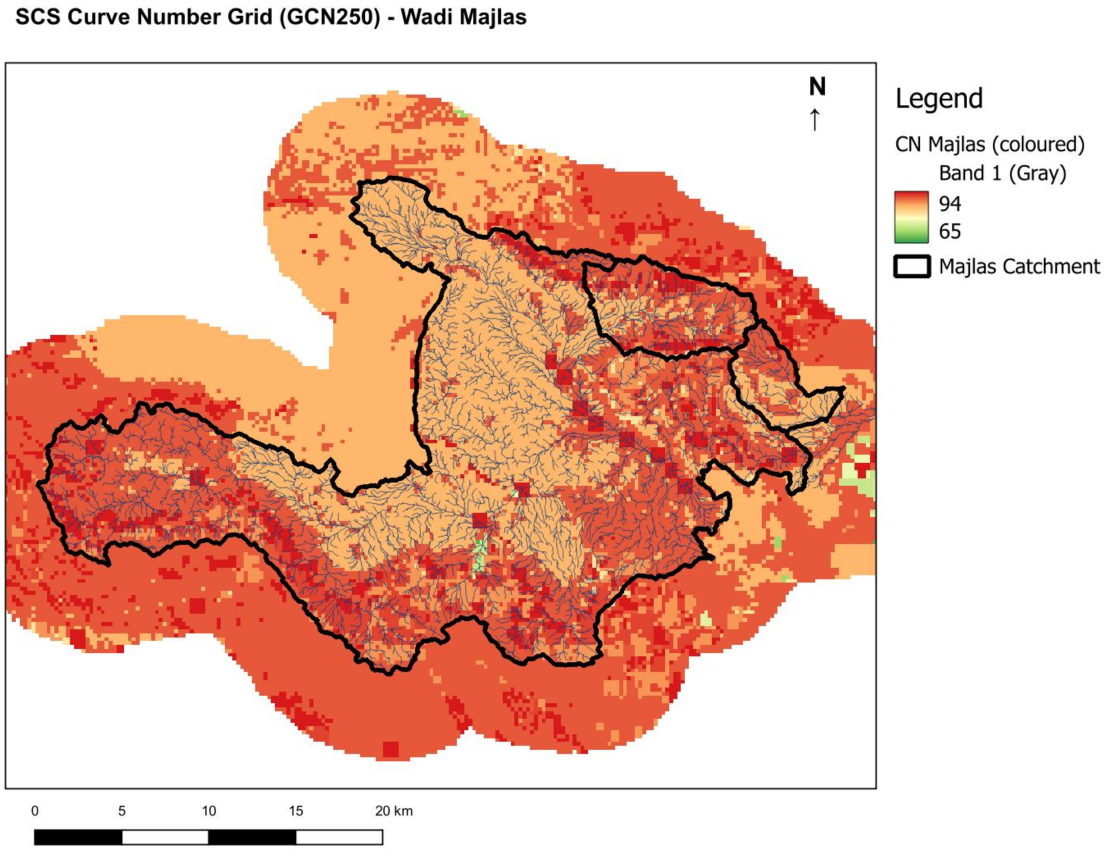


**Figure 9 SCS curve-number grid (GCN250) over the Wadi Majlas catchment (QGIS).**

## 7.3. AREA REDUCTION FACTOR

Point rainfall over-estimates the average depth over a large catchment; an Area Reduction Factor (ARF) from the extreme Oman storms was applied through the depth-area analysis in the final model:

$$ARF = 1 - 0.001\, A^{(1.25 - 0.062\, ln\, A)}$$

**Table 8 Area reduction factor applied in the depth-area analysis.**

| Area (km2) | Area reduction factor |
|---|---|
| 1 | 1.000 |
| 5 | 0.994 |
| 10 | 0.987 |
| 20 | 0.976 |
| 30 | 0.965 |
| 40 | 0.956 |
| 50 | 0.948 |

| Area (km2) | Area reduction factor |
|---|---|
| 100 | 0.913 |
| 200 | 0.864 |
| 500 | 0.776 |
| 1000 | 0.694 |
| 2000 | 0.606 |

## 7.4. CATCHMENT, STREAMS AND SUB-BASINS

Catchment and stream delineation used the 5 m DTM after sink-filling, flow-direction and flow-accumulation processing with a 1 km2 minimum drainage-area threshold. The HEC-HMS sub-basin delineation is shown in Figure 10. Sub-basin lag times were computed from the Kirpich formula and initial abstractions from the curve number:

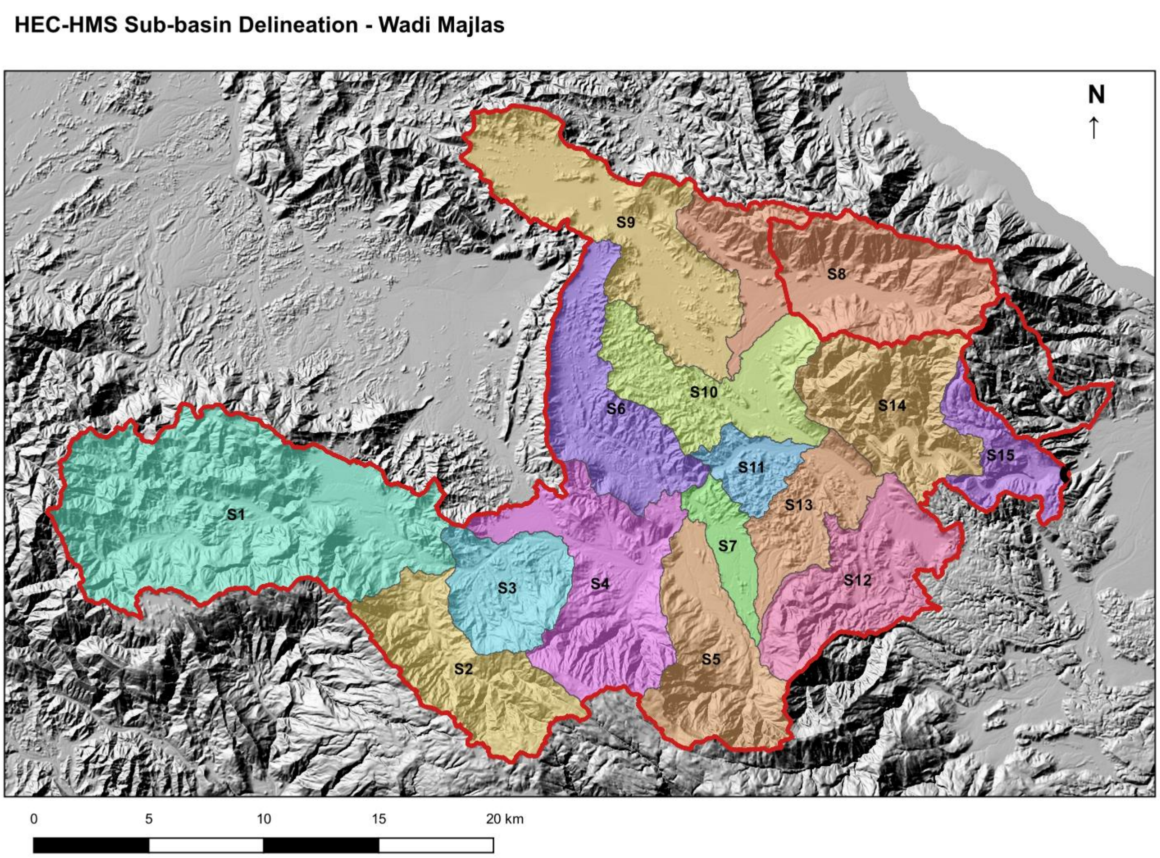


**Figure 10 HEC-HMS sub-basin delineation of the Wadi Majlas model (QGIS).**

$$t_{lag} = 0.6\,\frac{1}{52}\frac{(L_{10-85} \times 1000)^{0.77}}{S^{0.385}}$$

where L(10-85) is the 10-85% flow-path length (km) and S its slope. The 20 model sub-basins - 15 upstream of the dam plus the downstream sub-basins for the dyked channel - are summarised in Table 9.

**Table 9 Characteristics of the HEC-HMS sub-basins (Wadi Majlas model).**

| Sub-basin | Area (km2) | CN | Ia (mm) | Lag (min) | Downstream |
|---|---|---|---|---|---|
| S1 | 106.8 | 90.3 | 5.45 | 110 | J1 |
| S2 | 36.6 | 91 | 5.04 | 34.8 | J1 |
| S3 | 24.2 | 87.7 | 7.12 | 44 | J2 |
| S4 | 47.9 | 88 | 6.94 | 80.4 | J3 |
| S5 | 36.8 | 90 | 5.63 | 46.9 | J3 |
| S6 | 36.7 | 86.5 | 7.94 | 113 | J4 |
| S7 | 10.7 | 86.8 | 7.76 | 60.8 | J4 |
| S9 | 56.5 | 86.5 | 7.92 | 147 | J5 |
| S10 | 34.2 | 87.7 | 7.13 | 92.3 | J6 |
| S11 | 10.6 | 87.6 | 7.21 | 46.2 | J6 |
| S12 | 37.0 | 90.5 | 5.36 | 65.5 | J7 |
| S13 | 21.6 | 89.8 | 5.78 | 64.6 | J7 |
| S14 | 37.3 | 90 | 5.62 | 70.6 | J8 |
| S15 | 15.9 | 89.4 | 6.05 | 30.3 | Majlas Dam |
| Subbasin-1 | 40.4 | 90.1 | 5.55 | 43.2 | J9 |
| Subbasin-2 | 18.0 | 89.2 | 6.17 | 43.5 | J5 |
| Sdn1 | 3.0 | 88 | 6.92 | 28.1 | J10 |
| Sdn2 | 20.3 | 88.5 | 6.58 | 24.9 | J11 |
| Sdn3 | 3.5 | 88.6 | 6.52 | 25.7 | O1 |
| Sdn4 | 6.2 | 88.4 | 6.65 | 33.6 | O1 |
| TOTAL | 604.3 | | | | |

# 8. MODEL CALIBRATION AND REVIEW OF THE PREVIOUS STUDY

The model was calibrated against observed hydrographs from Cyclone Gonu (June 2007) - the largest instrumented event in the region - and checked against Cyclone Phet (2010). The previous study calibrated the model to observed discharge hydrographs at the two gauges near the dam, Hayfadh and Majlass. The recorded 24-hour Gonu depths at the contributing gauges are listed in Table 10.

On re-calibration, the Hayfadh gauge hydrograph was reproduced well with consistent, physically reasonable parameters (as also reported previously), but the observed hydrograph at the Majlass gauge could not be matched with the gauged Gonu rainfall. A sensitivity analysis progressively reduced the recorded storm rainfall to find the reduction needed to reproduce that hydrograph. Matching it required reducing the gauged 24-hour rainfall to physically implausible levels at several stations - for example 384 to 38 mm at Buei and 231 to 23 mm at Qurayat - while others required almost no change (Table 10).

**Table 10 Rainfall reduction required to reproduce the questioned Majlass observed hydrograph (Gonu 2007).**

| Gauge | Gonu 24-h (mm) | Reduced to match (mm) | Reduction |
|---|---|---|---|
| Jabal Abyadh East | 629 | 252 | -60% |
| Buei near Buei | 384 | 38 | -90% |
| Taba at Taba | 518 | 414 | -20% |
| Hayfadh at Hayfadh | 625 | 625 | 0% |
| Qurayat at Qurayat | 231 | 23 | -90% |

Such non-uniform, extreme reductions have no physical basis. The conclusion is that the **observed hydrograph at the Majlass gauge is erroneous** - most likely affected by a rating or gauging error at high stage - and should not have been used as a calibration target; this was not identified in the previous report. The present model was calibrated to the reliable Hayfadh hydrograph and to observed peak flows, validated against Cyclone Phet, and the questioned Majlass record was excluded. The adopted parameters are the GCN250 curve numbers (catchment average about 89) and Kirpich lag times of Table 9.

# 9. DESIGN FLOOD RESULTS

Design floods were computed for return periods from 2 to 10,000 years and the PMP, using the Flood-Mapping design rainfall and the calibrated model, in two configurations: **without** reservoir storage (dam sites as flow nodes: Majlas and Haifaz), and **with** the Majlas reservoir storage and spillway routed.

The Majlas reservoir has a full-supply level of 83 m and crest of 94 m, giving about 94 Mm3 of storage (dam height about 74.6 m); its elevation-area-storage relationship is shown in Figure 11.

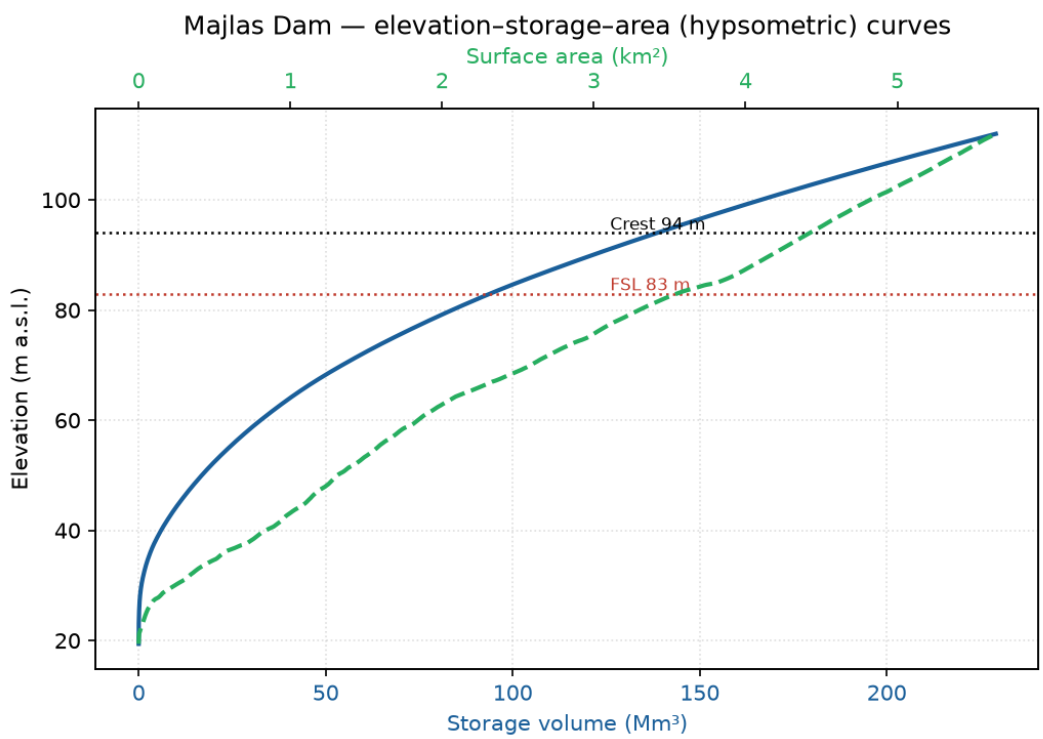


**Figure 11 Majlas Dam elevation-storage-area (hypsometric) curves.**

## 9.1. PEAK DISCHARGES

Table 11 gives peak discharges at key locations; Figure 12 shows the outlet peak-frequency curve for both configurations.

**Table 11 Design peak discharges (m3/s) at key locations, with and without the Majlas reservoir.**

| Location / configuration | 2-yr | 10-yr | 100-yr | 1,000-yr | 10,000-yr | PMP |
|---|---|---|---|---|---|---|
| Majlas Dam (inflow) - no res. | 92 | 1,161 | 3,452 | 8,289 | 14,166 | 18,726 |

| Location / configuration | 2-yr | 10-yr | 100-yr | 1,000-yr | 10,000-yr | PMP |
|---|---|---|---|---|---|---|
| Catchment outlet O1 - no res. | 86 | 1,126 | 3,351 | 8,026 | 13,798 | 18,692 |
| Downstream J10 - no res. | 89 | 1,144 | 3,398 | 8,131 | 13,923 | 18,532 |
| Downstream J12 - no res. | 88 | 1,135 | 3,394 | 8,139 | 13,968 | 18,751 |
| Majlas Dam (routed out) - with res. | 0 | 0 | 0 | 2,228 | 8,749 | 17,224 |
| Catchment outlet O1 - with res. | 17 | 216 | 449 | 2,180 | 8,585 | 17,286 |
| Downstream J10 - with res. | 2 | 29 | 56 | 2,211 | 8,665 | 17,123 |
| Downstream J12 - with res. | 17 | 221 | 430 | 2,198 | 8,664 | 17,303 |

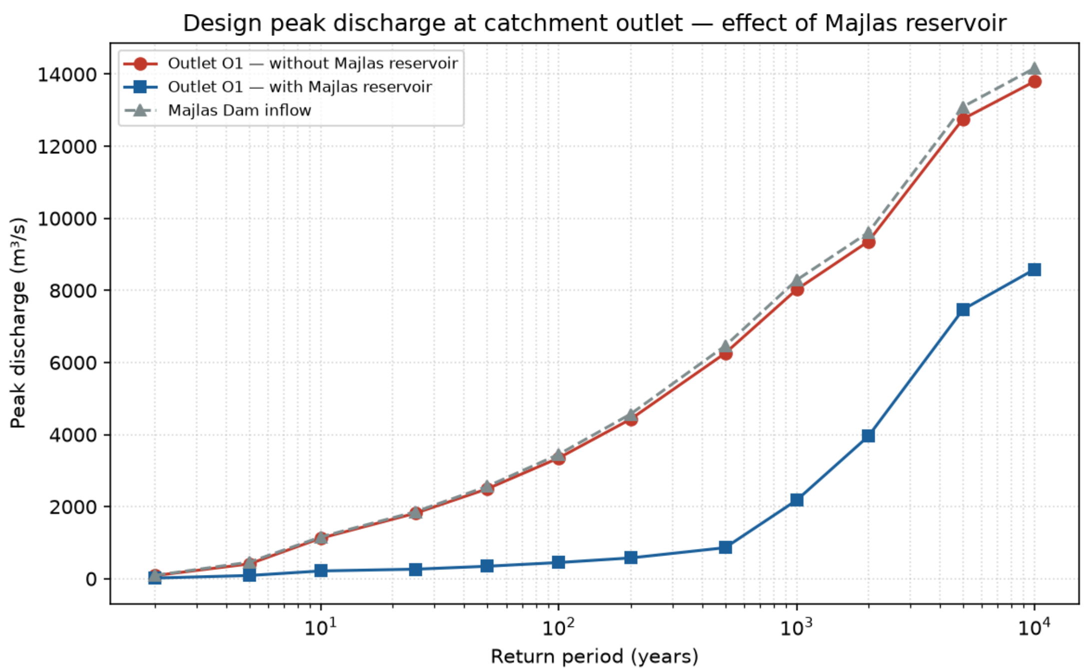


**Figure 12 Peak-discharge frequency at the catchment outlet - effect of the Majlas reservoir.**

## 9.2. RESERVOIR ROUTING AND ATTENUATION

The reservoir stores the full inflow up to about the 200-year event (zero spillway outflow) and strongly attenuates larger floods (Table 12, Figures 13-14). The 10,000-year peak is reduced from 14,166 to 8,749 m3/s (38%); even for the PMP the routed outflow (17,224 m3/s) stays below the inflow (18,726 m3/s).

**Table 12 Majlas reservoir routing - peak inflow, routed outflow and attenuation.**

| Return period | Peak inflow (m3/s) | Peak outflow (m3/s) | Attenuation |
|---|---|---|---|
| 2-yr | 92 | 0 | 100% |

| Return period | Peak inflow (m3/s) | Peak outflow (m3/s) | Attenuation |
|---|---|---|---|
| 5-yr | 460 | 0 | 100% |
| 10-yr | 1,161 | 0 | 100% |
| 25-yr | 1,859 | 0 | 100% |
| 50-yr | 2,561 | 0 | 100% |
| 100-yr | 3,452 | 0 | 100% |
| 200-yr | 4,566 | 0 | 100% |
| 500-yr | 6,459 | 877 | 86% |
| 1000-yr | 8,289 | 2,228 | 73% |
| 2000-yr | 9,607 | 4,029 | 58% |
| 5000-yr | 13,095 | 7,608 | 42% |
| 10000-yr | 14,166 | 8,749 | 38% |
| PMP | 18,726 | 17,224 | 8% |

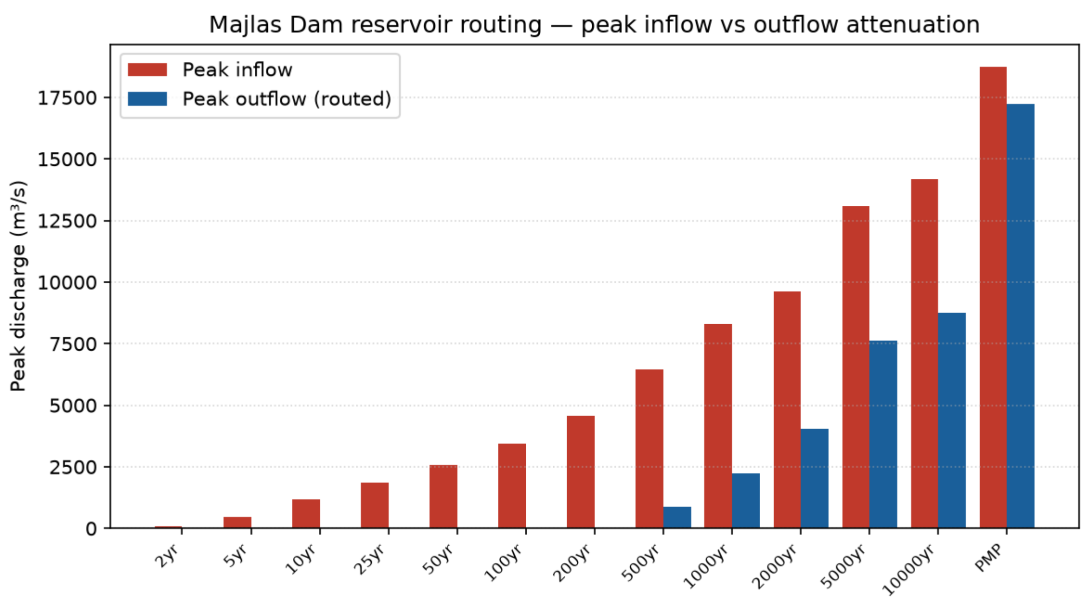


**Figure 13 Majlas reservoir routing - peak inflow versus routed outflow.**

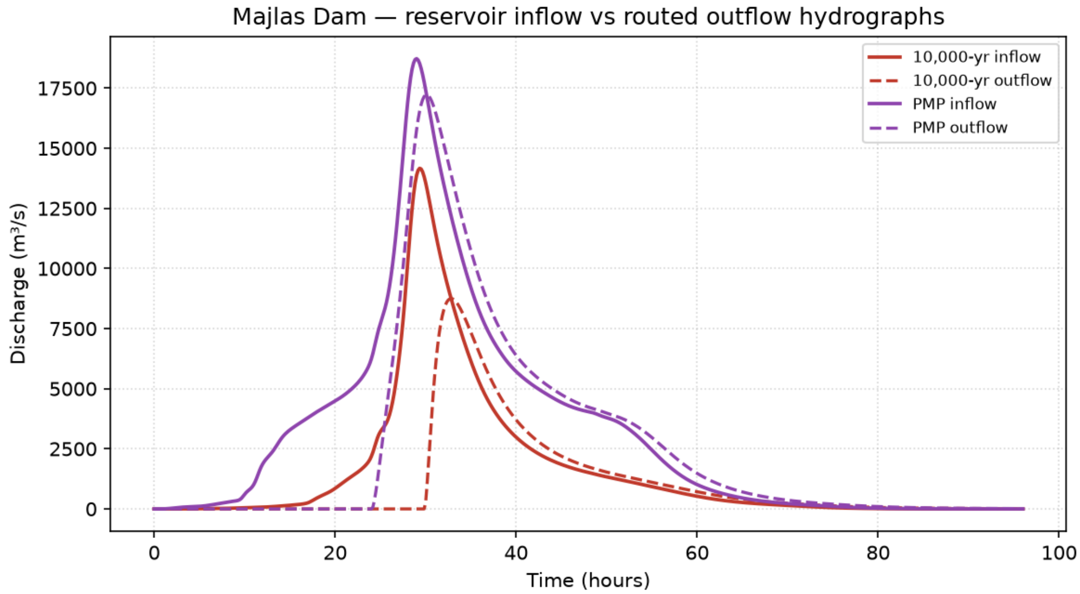


**Figure 14 Majlas reservoir inflow and routed-outflow hydrographs (10,000-year and PMP).**

The downstream peaks (J10, J12 and outlet O1 in Table 11) are the design flows for the dyked channel. Because the reservoir removes the upstream contribution up to moderate return periods, the downstream design flows under the with-reservoir configuration are governed by the local downstream sub-catchments to about the 200-year event and by reservoir spill for rarer events.

# 10. CONCLUSIONS AND RECOMMENDATIONS

- The catchment to the dam is 571 km2 (604 km2 with the downstream sub-catchments), mountainous and steep, with a catchment-average curve number of about 89.
- Annual maxima (maximum annual rolling sums) were extracted per water year; gauge-offline and clearly missed-storm years were excluded, and genuine 10-minute gaps filled by 3D-IDW in a single pass without reusing filled values or treating real zeros as gaps. The 2007 Cyclone Gonu maxima are retained as genuine extremes; no significant trend or change-point was found.
- The five fitting gauges form one statistically homogeneous region (24-hour H1 = -1.6; no discordant sites). Because Oman rainfall is moisture-limited (PMP ~ 1,300 mm), the GEV shape was constrained to the bounded case (kappa >= 0); the local index-flood 24-hour depths are ~1.5 times the Oman Flood-Mapping values at moderate return periods, converge with them at rare return periods and stay below the PMP. The Flood-Mapping depths were adopted for design.
- The observed Majlass Gonu hydrograph used for calibration in the previous study is erroneous (reproducible only with implausible rainfall reductions) and was excluded; this was not flagged previously.
- The Majlas reservoir stores the full inflow to about the 200-year event and attenuates the 10,000-year peak by about 38% (14,166 to 8,749 m3/s). Design peaks and hydrographs (2-10,000 years and PMP) are provided for the dam spillway and the downstream dyked channel.

It is recommended that design proceed on the Flood-Mapping-based design floods with reservoir routing, that the questioned historical gauge record be formally set aside, and that the local rainfall analysis be revisited as additional years of record accumulate.

# 11. REFERENCES

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